STAT3888 Visualisations and Tables for Question 2

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1 Tables

Table 1: Definitions and coding of variables used in the ALT and GGT analyses

Variable	Meaning	Type / coding
`ALT3`	Alanine aminotransferase (ALT) category	Categorical factor with ordered levels: Low / Borderline / High
`GGT3`	Gamma-glutamyl transferase (GGT) category	Categorical factor with ordered levels: Low / Borderline / High
`ExerciseGuidelines`	Whether recommended physical-activity guidelines were met	Binary factor: Yes / No
`Age`	Age in years	Continuous numeric
`Sex`	Sex	Binary factor: Female / Male
`Waist`	Waist circumference in cm	Continuous numeric
`AlcoholRisk`	Alcohol consumption risk level	Binary factor: Low risk / Medium–High risk
`SmokingDuration`	Duration of smoking history	Ordered categorical factor, grouped by years smoked
`NonSchoolQuals`	Whether a non-school educational qualification was completed	Binary factor: Yes / No
`IncomeDecile`	Income decile	Ordered categorical factor: deciles 1–10

Table 2: Distribution of exercise-guideline status and ALT and GGT outcome categories

Variable	Class	Count	Proportion (%)
`ExerciseGuidelines`	Yes	2464	51.7
	No	2298	48.3
`ALT3`	Low	3658	76.8
	Borderline	525	11.0
	High	579	12.2
`GGT3`	Low	3870	81.3
	Borderline	268	5.6
	High	624	13.1

Table 3: Summary of adjustment variables used in the ALT and GGT models. Numerical variables are reported as mean (IQR), with the IQR displayed as Q_1–Q_3; categorical variables are reported as count and proportion.

Variable	Mean (IQR)
Age	51.8 (37.0–62.0)
Waist	94.4 (83.0–104.0)

	Class	Count	Proportion (%)
AlcoholRisk	Low risk	4155	87.3
	Medium/High risk	607	12.7
SmokingDuration	Never smoked daily	2538	53.3
	Less than 10 years	512	10.8
	10–19 years	556	11.7
	20–29 years	479	10.1
	30–39 years	352	7.4
	40–49 years	238	5.0
	50 years and over	87	1.8
Sex	Female	2519	52.9
	Male	2243	47.1
NonSchoolQuals	Yes	3081	64.7
	No	1681	35.3
IncomeDecile	1	301	6.3
	2	473	9.9
	3	539	11.3
	4	560	11.8
	5	448	9.4
	6	448	9.4
	7	478	10.0
	8	482	10.1
	9	546	11.5
	10	487	10.2

Table 4: Association between exercise-guideline status and ALT and GGT categories

		Model 1		Model 2	
Biomarker	**Category**	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**
ALT	Low	Ref.	–	Ref.	–
	Borderline	0.92 (0.76, 1.10)	0.3495	0.89 (0.73, 1.08)	0.2289
	High	1.30 (1.09, 1.55)	0.0036 **	1.11 (0.92, 1.34)	0.2727
GGT	Low	Ref.	–	Ref.	–
	Borderline	1.27 (0.99, 1.63)	0.0549	1.13 (0.88, 1.47)	0.3409
	High	1.71 (1.44, 2.03)	< 0.0001 ****	1.35 (1.12, 1.62)	0.0014 **

Note. Odds ratios compare participants who did not meet exercise guidelines with those who did meet exercise guidelines. Low ALT/GGT is the reference outcome category. Model 1 includes exercise-guideline status only. Model 2 adjusts for sex, alcohol risk, smoking duration, non-school qualifications, income decile, age, and waist circumference. Significance codes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

2 Figures

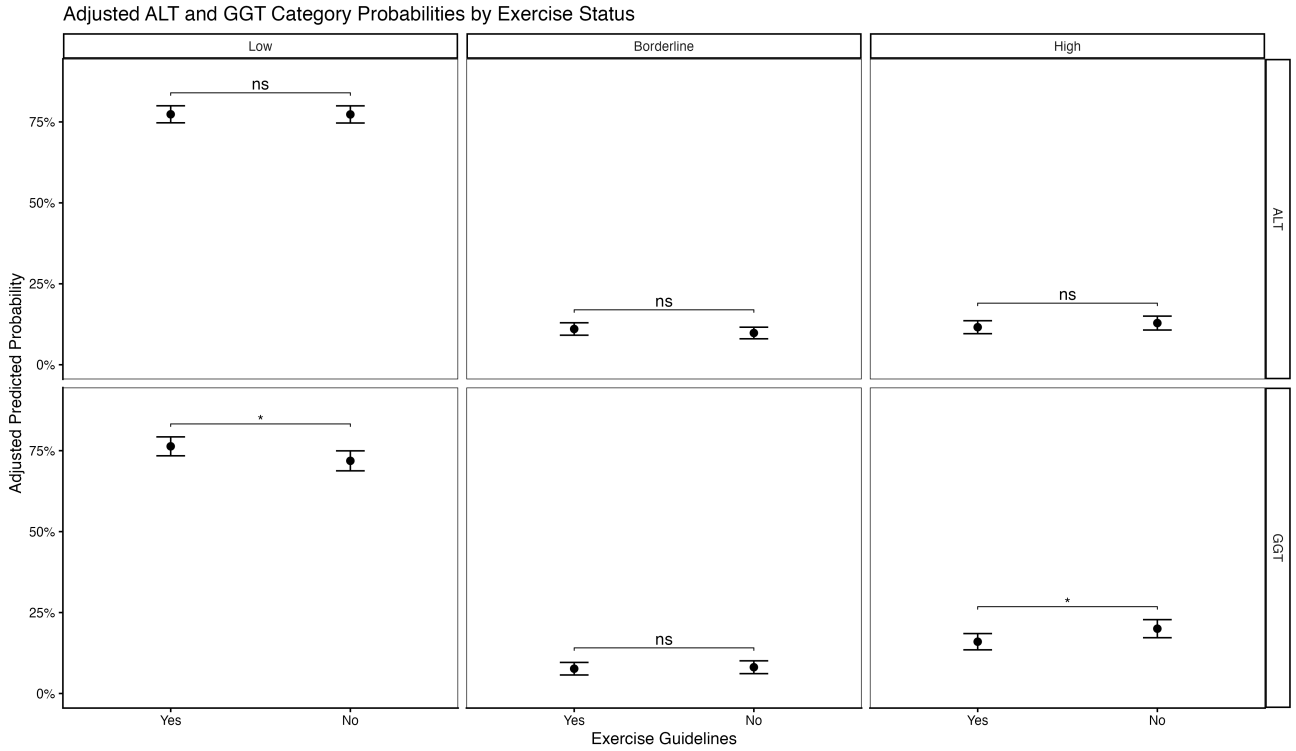


Figure 1: Adjusted predicted probabilities of low, borderline, and high ALT and GGT categories by exercise-guideline status. Points represent model-adjusted predicted probabilities from the multinomial logistic regression models, with error bars showing 95% confidence intervals. Brackets compare participants who met versus did not meet exercise guidelines within each biomarker category. Significance codes are based on Holm-adjusted pairwise comparisons.

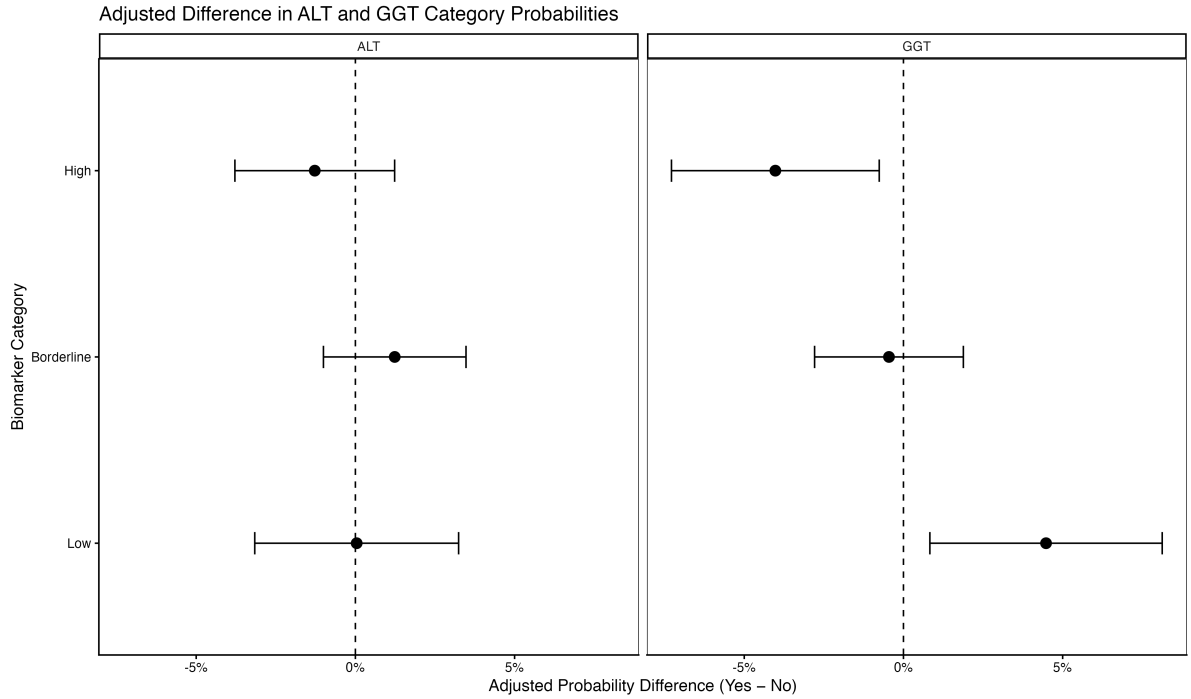


Figure 2: Adjusted differences in predicted ALT and GGT category probabilities between participants who met and did not meet exercise guidelines. Points show the estimated probability difference, defined as Exercise Yes minus Exercise No, with 95% confidence intervals. Positive values indicate a higher adjusted probability among participants meeting exercise guidelines, while negative values indicate a lower adjusted probability. The vertical dashed line at zero represents no difference between exercise groups.

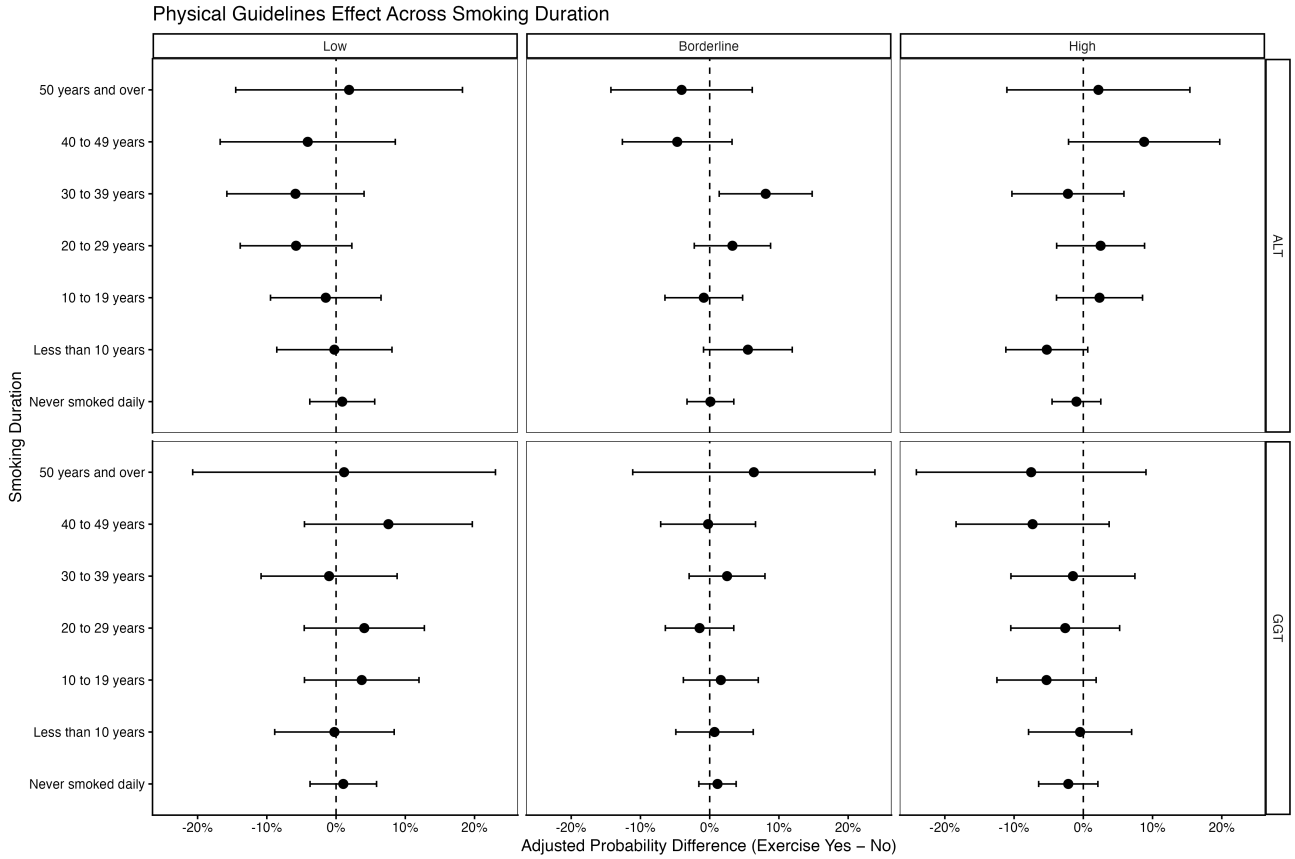


Figure 3: Adjusted exercise-associated differences in ALT and GGT category probabilities across smoking-duration groups. Within each smoking-duration category, points show the adjusted probability difference between participants who met and did not meet exercise guidelines (Exercise Yes minus Exercise No), with 95% confidence intervals. The vertical dashed line at zero denotes no exercise-associated difference. Variation in the estimated differences across smoking-duration categories illustrates potential effect modification by smoking duration.

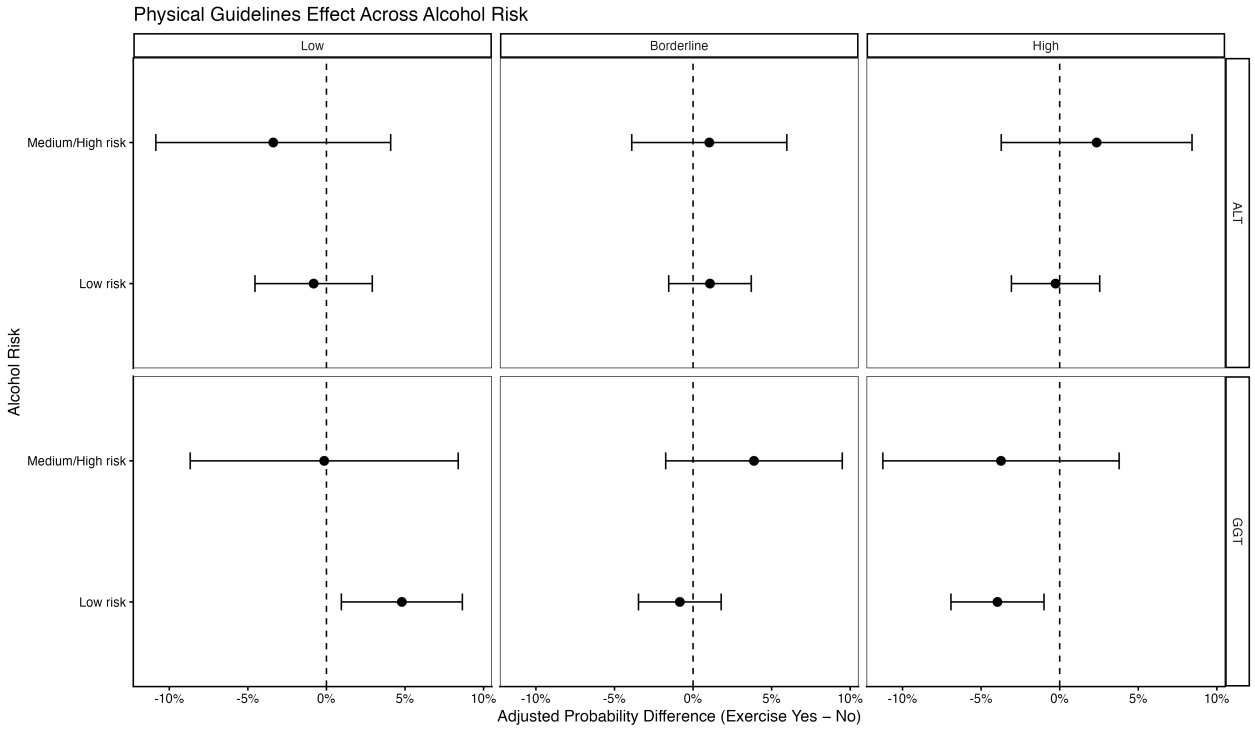


Figure 4: Adjusted exercise-associated differences in ALT and GGT category probabilities across alcohol-risk groups. Within each alcohol-risk category, points show the adjusted probability difference between participants who met and did not meet exercise guidelines (Exercise Yes minus Exercise No), with 95% confidence intervals. The vertical dashed line at zero denotes no exercise-associated difference. Differences in the estimated effects across alcohol-risk groups illustrate potential effect modification by alcohol consumption risk.